

Notes: Freund & Özden (AER, 2008)

Trade Policy and Loss Aversion

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1 Big picture

- **Question.** Why are declining industries so often protected, even though standard political economy models predict more protection for large or expanding sectors?
- **Core idea.** Trade policy is shaped not only by lobbying and industry size, but also by reference-dependent preferences and loss aversion. Losses relative to a reference point loom larger than gains.
- **Main theoretical contribution.** The paper embeds loss aversion into a Grossman–Helpman-style lobbying model of trade policy.
- **Key mechanism.** When a negative world price shock pushes an industry below its reference income/price, the government places extra weight on preventing losses. This can make protection rise after adverse shocks.
- **Important contrast.** In standard lobbying models, protection is increasing in the size of the domestic industry. In this model, protection can rise when an industry starts declining because preventing losses becomes politically valuable.
- **Main prediction 1: compensating protection.** For an intermediate range of low world prices, trade policy exactly offsets world price declines and keeps the domestic price fixed at the reference price.
- **Main prediction 2: hump-shaped protection.** As the world price falls, protection first rises, then reaches a compensating region, and then eventually declines when the domestic industry becomes too small and uncompetitive.
- **Main prediction 3: persistence.** Once protection is introduced, it can become part of the reference point, making removal politically difficult.
- **Empirical case.** The paper studies U.S. hot-rolled steel from 1981 to 2003.
- **Evidence.** In early low-price periods, changes in U.S. steel prices are weakly correlated with changes in world prices, consistent with compensating protection. In the final low-price period, correlation returns because the industry has shrunk and protection no longer fully offsets world price movements.
- **Economic takeaway.** Loss aversion gives microfoundations for the political tendency to protect declining sectors and explains why contingent protection instruments such as antidumping, quotas, and price floors are common.

2 Data and measurement

2.1 Theoretical primitives

- The model is a small open economy, so domestic trade policy does not affect world prices.
- There are $n + 1$ goods. Good 0 is the numeraire and is produced only with labor.
- Each non-numeraire good uses labor and a sector-specific factor.
- Owners of specific factors lobby for protection because tariffs raise domestic prices and factor returns.
- Consumers have standard consumption utility plus a behavioral loss-aversion term.

Interpretation: The model keeps the standard specific-factors political economy structure, then changes the welfare function by adding reference dependence and loss aversion.

2.2 Reference dependence and loss aversion

The paper assumes preferences have three behavioral features:

- **Reference dependence:** welfare depends on deviations from a reference utility or reference income.
- **Loss aversion:** losses below the reference point matter more than gains above it.
- **Diminishing sensitivity:** marginal disutility falls as the size of the loss grows.

Interpretation: The most important effect is local. When an industry is just below its reference point, marginal political pressure to avoid the loss is very strong.

2.3 Reference price

For sector i , the return to the specific factor is $\pi_i(p_i)$, and the reference return corresponds to a reference domestic price $\bar{p}_i$.

- If $p_i \geq \bar{p}_i$, the sector is not in the loss domain.
- If $p_i < \bar{p}_i$, sector-specific factor owners experience loss aversion.

Interpretation: A world price decline is politically important only when it threatens to push domestic factor returns below the reference point.

2.4 Empirical steel data

- The empirical section focuses on U.S. hot-rolled steel from 1981 to 2003.
- U.S. and world prices are monthly spot prices from World Steel Dynamics and Purchasing Magazine.
- The paper uses quarterly averages because price transmission may occur with a delay.
- The world price is the European export price quote.
- U.S. prices are average spot prices in four regions: East, South, Midwest, and West.
- The reference world price is the World Steel Dynamics “average times” price of \$290 per ton.
- Three low-price periods are identified: 1982–1987, 1991–1994, and 1998–2002.

Interpretation: The empirical test asks whether U.S. domestic prices stop tracking world prices during low-price periods, as compensating protection predicts.

3 Models and empirical specifications

3.1 Individual utility with loss aversion

A representative individual consumes the numeraire x_0 and differentiated goods x_i . Utility is standard consumption utility plus a loss-aversion term:

$$U = x_0 + \sum_{i=1}^n u_i(x_i) - Ih \left(\bar{U} - x_0 - \sum_{i=1}^n u_i(x_i) \right), \quad (1)$$

where $I = 1$ if utility is below the reference level and $I = 0$ otherwise.

Interpretation: The loss-aversion term is active only when the individual is below the reference point. The function $h(\cdot)$ is increasing and concave in the loss domain in the sense of diminishing sensitivity.

3.2 Indirect utility and the loss term

For an owner of sector-specific factor i , indirect utility is:

$$V_i(p) = E + s(p) - I_i h \left(\frac{\pi_i(\bar{p}_i) - \pi_i(p_i)}{\alpha_i N} \right). \quad (2)$$

The sectoral loss-aversion term is written:

$$H_i(p) = -I_i \alpha_i N h \left(\frac{\pi_i(\bar{p}_i) - \pi_i(p_i)}{\alpha_i N} \right). \quad (3)$$

Interpretation: The loss is tied to the fall in sector-specific income. The closer the sector is to the reference price, the stronger the marginal force from loss aversion.

3.3 Lobby welfare and aggregate welfare

Standard lobby welfare excluding loss aversion is:

$$W_i(p) = \pi_i(p_i) + \alpha_i N [r(p) + s(p)]. \quad (4)$$

Modified lobby welfare is:

$$G_i(p) = W_i(p) + H_i(p). \quad (5)$$

Aggregate welfare including loss aversion is:

$$G(p) = W(p) + H(p). \quad (6)$$

Interpretation: The model preserves the standard Grossman–Helpman welfare component but adds the loss-aversion distortion to both lobby welfare and social welfare.

3.4 Government objective

The government values campaign contributions and social welfare:

$$V = \sum_{i \in O} C_i(p) + aG(p), \quad (7)$$

where O is the set of organized lobbies and $a > 0$ is the weight on social welfare.

In equilibrium, the domestic price vector solves:

$$p^o = \arg \max_p \left\{ \sum_{i \in O} G_i(p) + aG(p) \right\}. \quad (8)$$

Interpretation: The equilibrium price maximizes the joint objective of organized lobbies and the government. Loss aversion changes this joint objective exactly when a sector is below its reference point.

3.5 Case 1: domestic price above the reference point

If $p_i^o > \bar{p}_i$, loss aversion is inactive. The equilibrium tariff has the standard Grossman–Helpman form:

$$\frac{t_i^o}{1 + t_i^o} = \left(\frac{I_{O_i} - \alpha_O}{a + \alpha_O} \right) \frac{z_i^o}{e_i^o}. \quad (9)$$

Interpretation: When the industry is not in the loss domain, the model collapses to the standard political contribution model.

3.6 Case 2: loss aversion with an interior solution

When $p_i^o < \bar{p}_i$ and the first-order condition is satisfied, the optimal trade tax becomes:

$$\frac{t_i^o}{1 + t_i^o} = \frac{(I_{Oi} - \alpha_O) + \left[(a + I_{Oi}) h_i' \left(\frac{\pi_i(\bar{p}_i) - \pi_i(p_i^o)}{\alpha_i N} \right) \right]}{a + \alpha_O} \frac{z_i^o}{e_i^o}. \quad (10)$$

Interpretation: The formula is the Grossman–Helpman tariff plus a behavioral term. The behavioral term raises protection when the industry is below its reference price.

3.7 Case 3: loss aversion with a corner solution

If the loss-aversion force is strong enough, the government sets the domestic price exactly at the reference price:

$$p_i^o = \bar{p}_i. \quad (11)$$

Interpretation: This generates compensating protection. A fall in the world price is exactly offset by a higher tariff so that the domestic price remains constant.

3.8 Comparative statics

The model predicts three price regions:

- **High world prices:** loss aversion is inactive and the model behaves like Grossman–Helpman.
- **Intermediate low world prices:** the domestic price is held at the reference price, so protection exactly offsets world price changes.
- **Very low world prices:** protection remains positive but can no longer fully compensate because the domestic industry shrinks and diminishing sensitivity weakens the behavioral effect.

Interpretation: Protection is hump-shaped in the world price decline: it rises, then exactly compensates, and finally declines as the industry becomes too small.

3.9 Empirical regression for U.S. steel

The empirical test is based on whether changes in U.S. prices respond to changes in world prices:

$$\Delta P_t^{US} = \beta_0 + \beta_1 \Delta P_t^{World} + \beta_2 (\Delta P_t^{World} \times Low1_t) + \beta_3 (\Delta P_t^{World} \times Low2_t) + \beta_4 (\Delta P_t^{World} \times Low3_t) + u_t. \quad (12)$$

- *Low1* is 1982–1987.
- *Low2* is 1991–1994.
- *Low3* is 1998–2002.

Interpretation: Under compensating protection, the sum $\beta_1 + \beta_j$ should be close to zero in low-price periods because domestic prices do not respond to world price movements.

4 Identification and empirical design

4.1 What identifies the steel evidence

- The model identifies periods in which world prices fall below a reference price.
- In those periods, compensating protection predicts low correlation between world and U.S. price changes.
- The paper compares low-price periods with high-price periods and with the full sample.
- It then tests whether domestic price changes respond to world price changes through correlations and regressions.

Interpretation: The steel evidence is a case-study test of the model’s comparative statics rather than a broad causal panel design.

4.2 Why the U.S. steel industry is useful

- Steel has a long history of lobbying and contingent protection.
- Hot-rolled steel is a major standardized product with available price data.
- The industry experienced repeated low-price episodes.
- Protection instruments such as trigger price mechanisms, voluntary restraint agreements, antidumping, and countervailing duties were used to stabilize domestic prices.

Interpretation: Steel is useful because the model predicts exactly the kind of contingent price-stabilizing protection observed in this sector.

4.3 What the evidence cannot fully prove

- The empirical section is a case study, not a full cross-industry test.
- The reference price is based on industry analysts’ “average times” benchmark.
- Other factors may affect U.S. and world steel prices.
- The model’s behavioral mechanism is inferred from policy patterns, not directly measured through individual preferences.

Interpretation: The steel evidence is best read as showing that observed policy patterns are consistent with the loss-aversion model and difficult to explain with standard models alone.

5 Tables and figures

This section keeps the compact *Read* plus *Economic message* format and directly inserts cropped visuals from the paper.

5.1 Figures 1 and 2: welfare with loss aversion

Read:

- Figure 1 shows the interior solution when the Grossman–Helpman price is below the reference price and loss aversion pulls the domestic price upward.
- The standard welfare component has a maximum at p_i^{GH} .

- The loss-aversion component is steep near the reference point and can dominate the standard welfare loss from protection.
- Figure 2 shows the corner solution where the domestic price is set exactly at the reference price $\bar{p}_i$.

Economic message: Loss aversion creates a new political force. The government may raise protection not because the industry is large, but because the industry is just below its reference point.

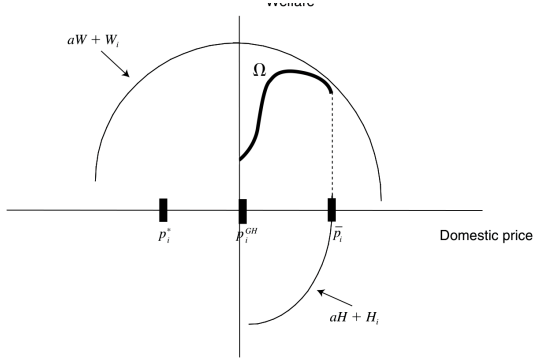


FIGURE 1. WELFARE FUNCTION AND LOSS AVERSION WITH INTERNAL SOLUTION

(a) Figure 1: internal solution.

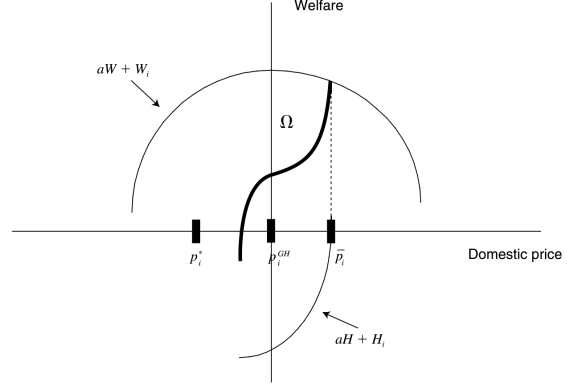


FIGURE 2. WELFARE FUNCTION AND LOSS AVERSION WITH CORNER SOLUTION

(b) Figure 2: corner solution.

5.2 Figures 3 and 4: compensating protection and hump-shaped tariffs

Read:

- Figure 3 plots equilibrium domestic prices against world prices.
- In the compensating-protection range, the domestic price is flat at the reference price even as world prices move.
- Figure 4 plots tariffs against world prices.
- Protection rises as the world price declines into the loss region, then eventually falls when the industry is too small and uncompetitive.

Economic message: The model predicts a nonmonotonic relationship between protection and world prices. This differs sharply from standard models that predict protection rises with domestic industry size.

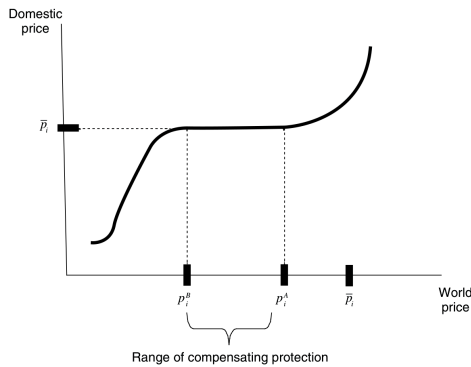


FIGURE 3. DOMESTIC AND WORLD PRICES

(a) Figure 3: domestic and world prices.

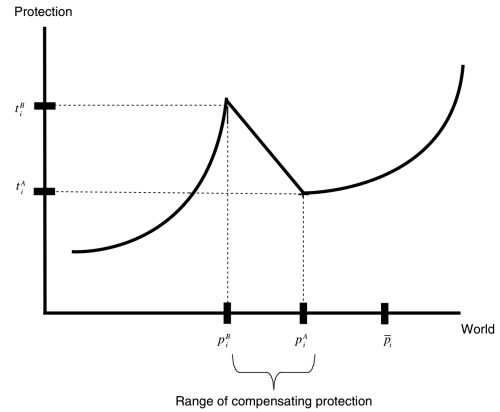


FIGURE 4. TARIFFS AND WORLD PRICES

(b) Figure 4: tariffs and world prices.

5.3 Figure 5: U.S. steel market data

Read:

- The upper panel compares U.S. and world prices for hot-rolled steel.
- The lower panel shows imports relative to domestic production capacity.
- In the 1980s and early 1990s, U.S. prices appear stabilized relative to world prices.
- In the late 1990s and early 2000s, import penetration rises sharply, and the industry becomes less able to sustain compensating protection.

Economic message: The figure visually matches the model: protection can stabilize prices in early low-price periods, but this ability weakens as the domestic industry declines.

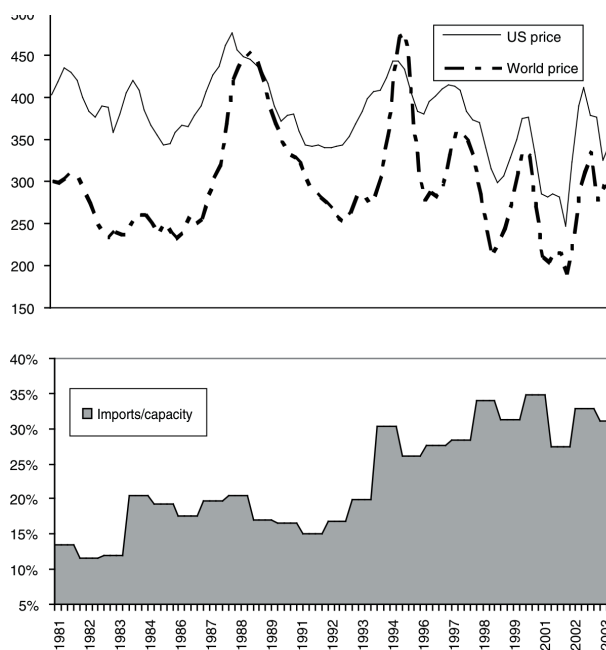


FIGURE 5. STEEL MARKET DATA

Source: Price data are for hot-rolled steel in \$US from World Steel Dynamics, Price Track (1981–2003). Employment, imports, and capacity are from AISI (1981–2003).

5.4 Tables 1 and 2: price correlations and regressions

Read:

- Table 1 shows that, for the full sample, changes in U.S. prices and world prices are positively correlated.
- In the first low-price period, 1982–1987, the correlation is only 0.15 and statistically insignificant.
- In the second low-price period, 1991–1994, the correlation is 0.34 and statistically insignificant.
- In the third low-price period, 1998–2002, the correlation is 0.81 and statistically significant.
- Table 2 shows that interaction terms for the first and second low-price periods reduce the effect of world price changes on U.S. prices.

- The tests cannot reject full compensating protection in the first two low-price periods, but they reject it in the third low-price period.

Economic message: The early low-price periods look like compensating protection: U.S. domestic prices stop moving with world prices. The final low-price period does not, consistent with the model’s prediction that protection weakens when the domestic industry shrinks.

TABLE 1—CORRELATION BETWEEN CHANGES IN US PRICES AND CHANGES IN WORLD PRICES

Quarterly data					
	Low price 1 1982–1987	High price 1987–1991	Low price 2 1991–1994	High price 1994–1998	Low price 3 1998–1999 2000–2002
Full sample					
0.62*** (0.00)	0.15 (0.52)	0.71*** (0.00)	0.34 (0.30)	0.79*** (0.00)	0.81*** (0.01)

Notes: Low price is world price below WSD “average times” price for hot-rolled steel for more than two consecutive quarters. Significance levels in parentheses.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.

(a) Table 1: correlations.

TABLE 2—THE EFFECT OF CHANGES IN WORLD PRICES ON DOMESTIC PRICES

	(1)	(2)
$\Delta \text{worldprice}$	0.41*** [5.40]	0.43*** [4.54]
$\Delta \text{worldprice} \times \text{low1}$		-0.30* [1.71]
$\Delta \text{worldprice} \times \text{low2}$		-0.24 [1.30]
$\Delta \text{worldprice} \times \text{low3}$		0.09 [0.57]
test A: $\Delta \text{worldprice} + \Delta \text{worldprice} \times \text{low1} = 0$		0.72 [0.40]
test B: $\Delta \text{worldprice} + \Delta \text{worldprice} \times \text{low2} = 0$		1.36 [0.25]
test C: $\Delta \text{worldprice} + \Delta \text{worldprice} \times \text{low3} = 0$		16.08 [0.00]
Observations	90	90
R-squared	0.39	0.41

Notes: Robust *t*-statistics in brackets. Regressions run with a constant not reported.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

(b) Table 2: regressions.

6 Short summary

- The paper explains why declining industries receive protection.
- Standard political economy models predict more protection for large and expanding sectors.
- Freund and Özden add loss aversion and reference dependence to a Grossman–Helpman lobbying model.
- When an industry is below its reference price, preventing losses becomes politically valuable.
- This creates compensating protection: tariffs exactly offset world price declines over an intermediate range.
- Protection is hump-shaped as world prices fall: it rises, then compensates, then eventually falls.
- Protection can become persistent because past protection enters the reference point.
- The U.S. steel case matches the model: early low-price periods show weak pass-through from world prices to U.S. prices.
- In the final low-price period, U.S. prices again track world prices, consistent with a smaller domestic industry and weaker protection.
- The main lesson is that behavioral preferences can explain contingent and declining-industry protection better than standard lobbying models alone.

7 Conclusion

7.1 Core conclusion

Freund and Özden show that loss aversion changes the political economy of trade policy. If governments care about preventing losses relative to a reference point, they may protect declining industries, even when standard lobbying models would predict less protection.

7.2 Economic intuition

The model balances two forces. The standard effect says that larger industries gain more from protection, so protection should rise with industry size. The behavioral effect says that industries just below their reference point generate strong political pressure because losses loom large. When the behavioral effect dominates, trade policy stabilizes domestic prices. When the industry declines too far, the standard effect dominates again and protection falls.

7.3 Takeaway

The paper provides a behavioral foundation for real-world contingent protection. Instruments such as antidumping, price floors, quotas, and voluntary restraint agreements often operate exactly as the model predicts: they bind when world prices fall and domestic firms face losses, but they become less effective when the protected industry becomes too small or too uncompetitive.